Unification and Resolution.

```
def unify(x, y, sub={}):
    if x == y:
        return sub
    if isinstance(x, str) and x.islower():
        sub[x] = y
        return sub
    if isinstance(y, str) and y.islower():
        sub[y] = x
        return sub
    if isinstance(x, list) and isinstance(y, list) and len(x) == len(y):
        for a, b in zip(x, y):
            sub = unify(a, b, sub)
            if sub is None:
                return None
        return sub
    return None
def negate(literal)
    return "~" + literal if not literal.startswith("~") else literal[1:]
print(unify(["P", "x"], ["P", "John"]))
print(negate("P"))
```

output: {'x': 'John'}

~P